ECO2011 Basic Microeconomics

Mankiw Chapter 1 (Ten Principles)

Mankiw Chapter 2 (Thinking Like an Economist)

2026 Spring

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Ice Breaking

- What's your name?
- Where do you come from?
- What do you know about economics/ Why do you want to study economics?

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Ten Principles of Economics

- Resources are scarce
- Scarcity: the limited nature of society's resources
  - Society has limited resources
    - Cannot produce all the goods and services people wish to have
- Economics
  - The study of how society manages its scarce resources

Ten Principles of Economics

- Economists study:
  - How people decide what to buy, how much to work, save, and spend
  - How firms decide how much to produce, how many workers to hire
  - How society decides how to divide its resources between national defense, consumer goods, protecting the environment, and other needs

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How People Make Decisions

- Decision making is at the heart of economics.
- The first four principles deal with how people make decisions.

Principle 1: People Face Trade-offs

- To get something that we like, we have to give up something else that we also like
  - Going to a party the night before an exam
    - Less time for studying
  - Having more money to buy stuff
    - Working longer hours, less time for leisure
  - Other examples?

Principle 1: People Face Trade-offs

- Society faces trade-offs:
  - The more it spends on national defense (guns) to protect its shores
    - The less it can spend on consumer goods (butter) to raise the standard of living at home
  - Pollution regulations: cleaner environment and improved health
    - But at the cost of reducing the incomes of the firms' owners, workers, and customers

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Principle 1: People Face Trade-offs

- Efficiency: society gets the most from its scarce resources
- Equality: prosperity is distributed uniformly among society's members
- Tradeoff:
  - To achieve greater equality, could redistribute income from wealthy to poor
  - But this reduces incentive to work and produce, shrinks the size of economic "pie"

Decisions require comparing costs and benefits of alternatives.

- Zhang San is a freshman who just entered CUHK(SZ).
  - At the beginning of the first semester, he needs to pick a school
  - Two nights before the ECO2011 midterm exam, he needs to decide whether to study for the exam or prepare for a club interview
  - At the end of the first semester, he needs to pick a major within his school
  - ……

Principle 2: The Cost of Something Is What You Give Up to Get It

- Making decisions:
  - Compare costs with benefits of alternatives
  - Opportunity costs
- Opportunity cost
  - Whatever must be given up to obtain some item

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Examples

- The opportunity cost of:
  - Going to college for a year
    - Tuition, books, and fees
    - PLUS foregone wages
  - Going to the movies
    - The price of the movie ticket
    - PLUS the value of the time you spend in the theater
  - Other Examples?

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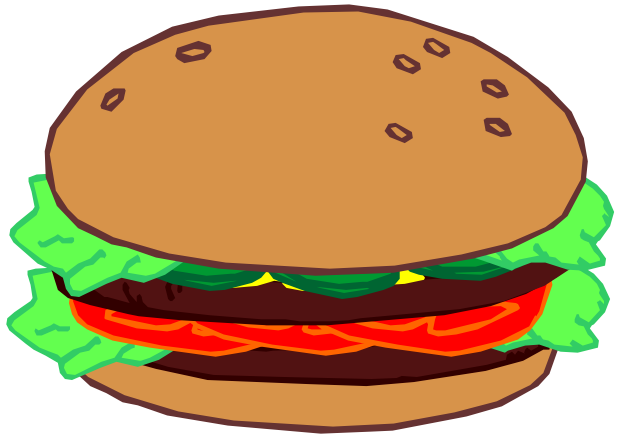
“There is no such thing as a free lunch!”

- Even a "free" lunch is not free.
  - Your friend says to you ``My treat!"
  - Is it really free?
  - It consumes your time, effort...



Zhang San's daily life continues

- Zhang San is a freshman who just entered CUHK(SZ).
  - He is at home on Sunday, trying to figure out how much time he would spend in studying economics.

Principle 3: Rational People Think at the Margin

- A person is **rational** if she systematically and purposefully does the best she can to achieve her objectives.
- Many decisions are not "all or nothing," but involve **marginal changes** – incremental adjustments to an existing plan.
- Evaluating the costs and benefits of marginal changes (MC and MB) is an important part of decision making.

  --The difference matters.

Examples

- Cell phone users with unlimited minutes (the minutes are free at the margin)
  - Are often prone to making long/frivolous calls
  - Marginal benefit of the call > 0
- A manager considers whether to increase output
  - Compares the cost of the needed labor and materials to the extra revenue

Principle 4: People Respond to Incentives

- Incentive
  - Something that induces a person to act
- Examples:
  - When gas prices rise, consumers buy more hybrid cars and fewer gas guzzling SUVs
  - When cigarette taxes increase, teen smoking falls
  - Other examples?

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Active Learning 1 Applying the principles

You are selling your 2020 Model 3. You have already spent $1,000 on repairs. At the last minute, the transmission dies. You can pay $900 to have it repaired, or sell the car "as is." In each of the following scenarios, should you have the transmission repaired? Explain.

A. Blue book value (what you could get for the car) is $7,500 if transmission works, $6,200 if it doesn't.

B. Blue book value is $6,300 if transmission works, $5,500 if it doesn't.

Active Learning 1 Applying the principles

Cost of fixing the transmission = \$900

A. Blue book value is \$7,500 if transmission works, \$6,200 if it doesn't

- Benefit of fixing transmission = \$1,300 (= 7500 – 6200)
- Get the transmission fixed

B. Blue book value is \$6,300 if transmission works, \$5,500 if it doesn't

- Benefit of fixing the transmission = \$800 (= 6300 – 5500)
- Do not pay \$900 to fix it

How People Interact

- An “economy” is just a group of people interacting with each other.
- The next three principles deal with how people interact.

Let's zoom in to Zhang San's life again

- Zhang San is a freshman who just entered CUHK(SZ)
  - He wears shoes from Anta, which he does not make himself
  - He eats noddle for lunch, but he does not know how to plant wheat
  - He reads Mankiw's textbook for ECO2011, but he does not know how to make papers
  - ……

Principle 5: Trade Can Make Everyone Better Off

- People benefit from trade:
  - People can buy a greater variety of goods and services at lower cost
- Countries benefit from trade and specialization
  - Get a better price abroad for goods they produce
  - Buy other goods more cheaply from abroad than could be produced at home

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Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

- Market
  - A group of buyers and sellers (need not be in a single location)
- "Organize economic activity" means determining
  - <u>what</u> goods to produce
  - <u>how</u> to produce them
  - <u>how much</u> of each to produce
  - <u>who</u> gets them

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Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

- In a market economy, these decisions result from the interactions of many households and firms.
- Famous insight by Adam Smith in The Wealth of Nations (1776):
  - Each of these households and firms acts as if "led by an invisible hand" to promote general economic well-being

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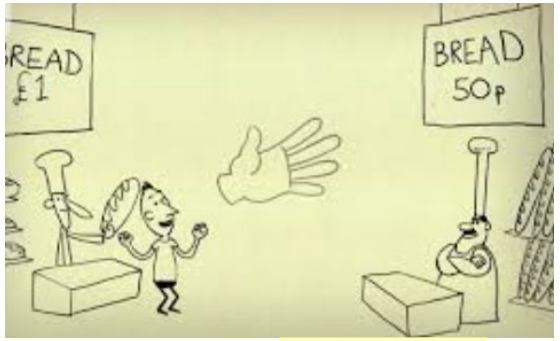
Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

- Prices:
  - Determined: interaction of buyers and sellers
  - Reflect the good's value to buyers
  - Reflect the cost of producing the good



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- Invisible hand:
  - Prices guide self-interested households and firms to make decisions that maximize society's economic well-being

Alright, Zhang San Again

- Zhang San is a freshman who just entered CUHK(SZ)
  - Zhang San is considering buying a bicycle, what will he do if he knows for sure the bicycle will got stolen?

Principle 7: Governments Can Sometimes Improve Market Outcomes

- Important role for govt: enforce property rights (with police, courts)
- People are less inclined to work, produce, invest, or purchase if large risk of their property being stolen.
  - A restaurant won't serve meals if customers do not pay before they leave.
  - A music company won't produce CDs if too many people avoid paying by making illegal copies.
  - Other examples?

Principle 7: Governments Can Sometimes Improve Market Outcomes

- Govt may alter market outcome to promote efficiency
- Market failure, when the market fails to allocate society's resources efficiently. Causes:
  - externalities, when the production or consumption of a good affects bystanders (examples?)
  - market power, a single buyer or seller has substantial influence on market price (e.g. monopoly)
- In such cases, public policy may increase efficiency.

Principle 7: Governments Can Sometimes Improve Market Outcomes

- Govt may alter market outcome to promote equity
- If the market's distribution of economic well-being is not desirable, tax or welfare policies can change how the economic "pie" is divided.

Active Learning 2 Discussion Question

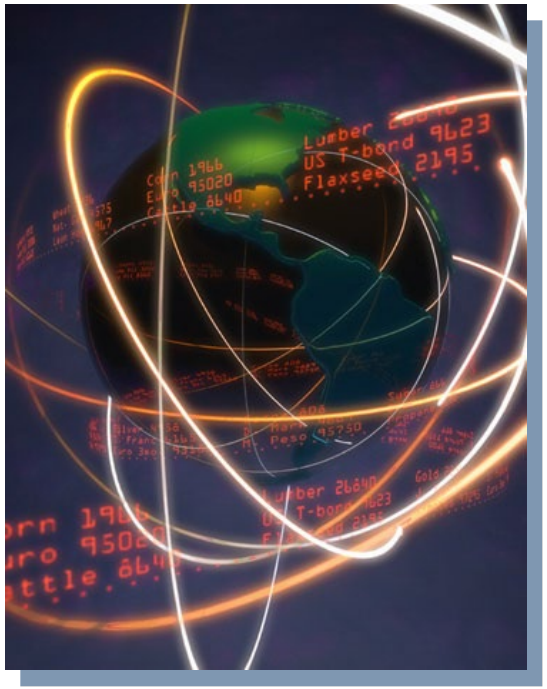
In each of the following situations, what is the government's role?

Does the government's intervention improve the outcome?

a. Workplace safety regulations
b. Public highways
c. Patent laws, which allow drug companies to charge high prices for life-saving drugs

How the economy as a whole works

- The last three principles deal with the economy as a whole.



Principle 8: Country's Standard of Living Depends on Its Ability to Produce Goods and Services

- Huge variation in living standards
  - Across countries and over time
  - Average income in rich countries
    - Is more than ten times average income in poor countries
  - The U.S. standard of living today
    - Is about eight times larger than 100 years ago

Principle 8: Country's Standard of Living Depends on Its Ability to Produce Goods and Services

- Productivity: most important determinant of living standards
  - Quantity of goods and services produced from each unit of labor input
  - Depends on the equipment, skills, and technology available to workers
  - Other factors (e.g., labor unions, competition from abroad) have far less impact on living standards

Principle 9: Prices Rise When the Government Prints Too Much Money

- Inflation
  - An increase in the overall level of prices in the economy
- In the long run
  - Inflation is almost always caused by excessive growth in the quantity of money, which causes the value of money to fall
  - The faster the government creates money, the greater the inflation rate

Principle 10: Society Faces a Short-run Trade-off between Inflation and Unemployment

- Short-run trade-off between unemployment and inflation
  - Over a period of a year or two, many economic policies push inflation and unemployment in opposite directions
  - More money → More spending and demand → Higher prices but more hiring (less unemployment)

What Have We Learned So Far?

- Fundamental lessons about individual decision making:
  - People face trade-offs among alternative goals
  - The cost of any action is measured in terms of forgone opportunities
  - Rational people make decisions by comparing marginal costs and marginal benefits
  - People change their behavior in response to the incentives they face

What Have We Learned So Far?

- Fundamental lessons about interactions among people:
  - Trade and interdependence can be mutually beneficial
  - Markets are usually a good way of coordinating economic activity among people
  - The government can potentially improve market outcomes by remedying a market failure or by promoting greater economic equality

What Have We Learned So Far?

- Fundamental lessons about the economy as a whole:
  - Productivity is the ultimate source of living standards
  - Growth in the quantity of money is the ultimate source of inflation
  - Society faces a short-run trade-off between inflation and unemployment

The Economist as a Scientist

- Economists play two roles:
  1. Scientists: try to explain the world
  2. Policy advisors: try to improve it
- As scientists, economists employ the scientific method
  - Dispassionate development and testing of theories about how the world works

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Assumptions & Models

- Assumptions simplify the complex world, make it easier to understand.
- Example: When studying international trade, we might assume the world consists of two countries and two goods.
  - Is it realistic?
  - Is it useful?
- Economists use models to study economic issues. A model is a highly simplified representation of a more complicated reality.

Microeconomics and Macroeconomics

- Microeconomics
  - The study of how households and firms make decisions and how they interact in markets
- Macroeconomics
  - The study of economy-wide phenomena, including inflation, unemployment, and economic growth

The Economist as Policy Advisor

- As scientists, economists make positive statements, which attempt to describe the world as it is.
- As policy advisors, economists make normative statements, which attempt to prescribe how the world should be.
- Positive statements can be confirmed or refuted, normative statements cannot.
- Govt employs many economists for policy advice.
  - E.g., the U.S. President has a Council of Economic Advisors, which Mankiw chaired from 2003 to 2005.

Active Learning 3 Positive vs. Normative

Which of these statements are "positive" and which are "normative"? How can you tell the difference?

a. Prices rise when the government increases the quantity of money.
b. The government should print less money.
c. A tax cut is needed to stimulate the economy.
d. An increase in the price of burritos will cause an increase in consumer demand for music downloads.

Active Learning 3 Answers

a. Prices rise when the government increases the quantity of money.

Positive – describes a relationship, could use data to confirm or refute.

b. The government should print less money.

Normative – this is a value judgment, cannot be confirmed or refuted.

Active Learning 3 Answers

c. A tax cut is needed to stimulate the economy.

Normative – another value judgment.

d. An increase in the price of burritos will cause an increase in consumer demand for music downloads

Positive – describes a relationship.

Note that a statement need not be true to be positive.

Why Economists Disagree

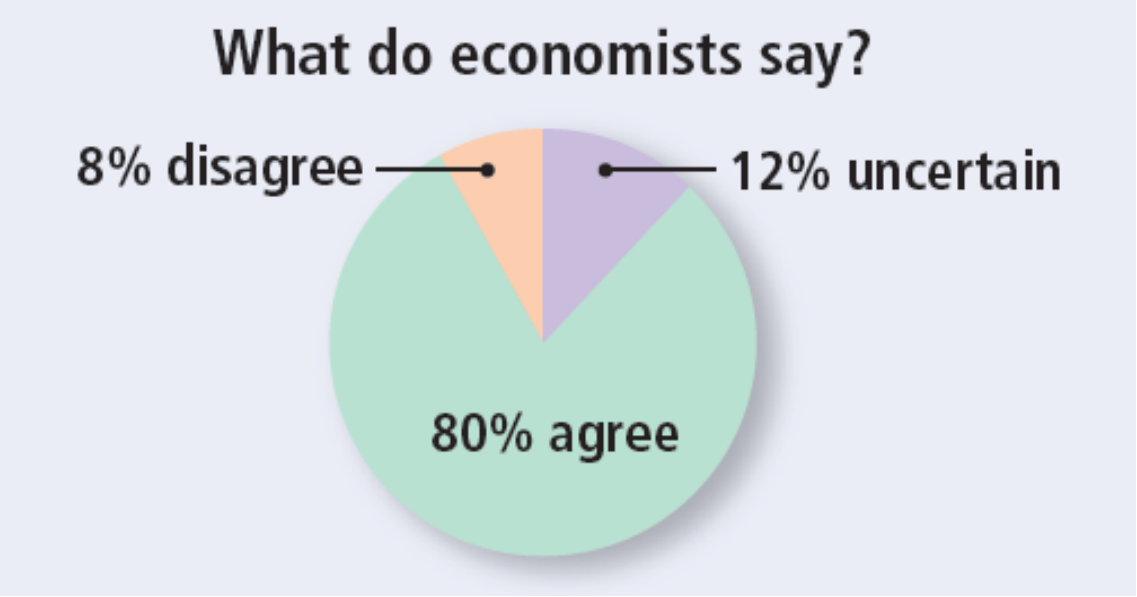
- Economists often give conflicting policy advice
  - Can disagree about the validity of alternative positive theories about the world
  - May have different values and, therefore, different normative views about what policy should try to accomplish
- Yet, there are many propositions about which most economists agree

Propositions about Which Most Economists Agree (and % agreeing)

- A ceiling on rents reduces the quantity and quality of housing available. (93%)
- Tariffs and import quotas usually reduce general economic welfare. (93%)
- A large federal budget deficit has an adverse effect on the economy. (83%)
- A minimum wage increases unemployment among young and unskilled workers. (79%)
- Effluent taxes and marketable pollution permits represent a better approach to pollution control than imposition of pollution ceilings. (78%)

Case: Ticket Resale

- "Laws that limit the resale of tickets for entertainment and sports events make potential audience members for those events worse off on average."



FYI: Who Studies Economics?

- Ronald Reagan, President of the United States
- Barbara Boxer, U.S. Senator
- Sandra Day-O'Connor, Supreme Court Justice
- Anthony Zinni, General, U.S. Marine Corps
- Kofi Annan, Secretary General, United Nations
- Meg Witman, Chief Executive Officer, eBay
- Steve Ballmer, Chief Executive Officer, Microsoft
- John Elway, NFL Quarterback
- Tiger Woods, Golfer
- Ben Stein, Political Speechwriter, Actor, Game Show Host
- Arnold Schwarzenegger, Governor of California, Actor
- Mick Jagger, Singer for the Rolling Stones

End